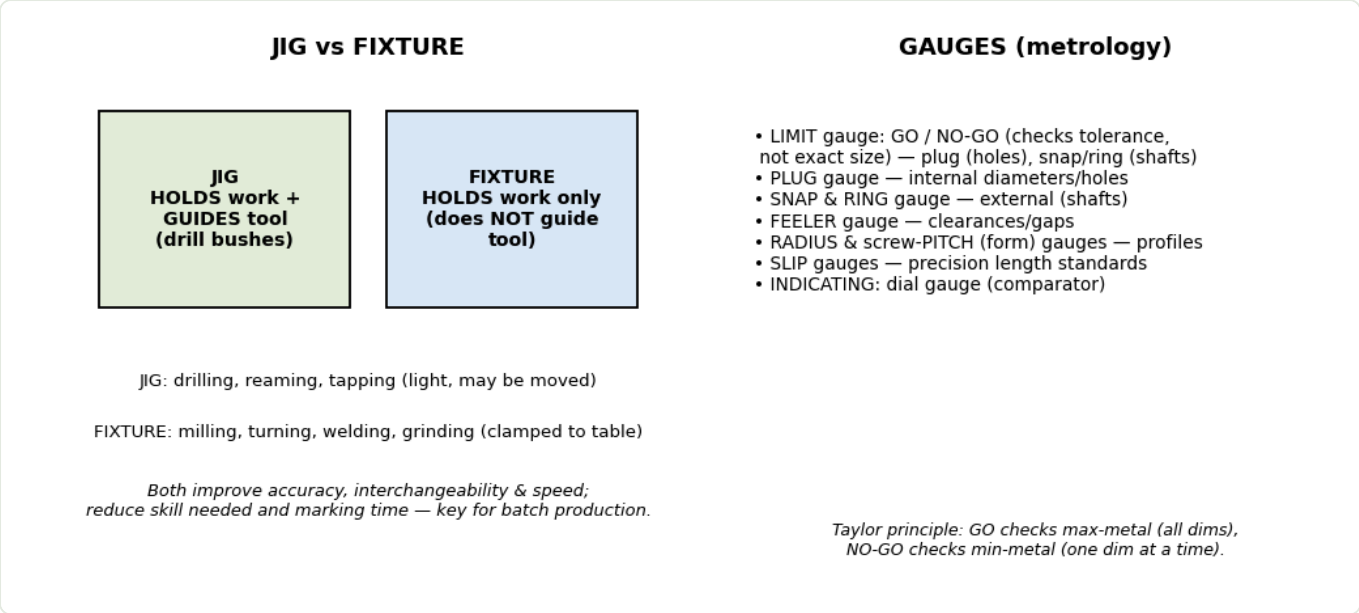
UNIT 8 · TOPIC 4(B)

Jigs, Fixtures and Gauges

Syllabus topic: Metal cutting and machining; jigs, fixtures and gauges



Jig vs fixture, and gauge types

Jigs vs fixtures

Both are work-holding devices that ensure accuracy and interchangeability in batch production, and cut marking/setting time and the skill needed.

	JIG	FIXTURE
Function	Holds the work AND GUIDES the cutting tool (through hardened drill BUSHES)	Only HOLDS and locates the work — does NOT guide the tool
Operations	DRILLING, reaming, tapping, boring	MILLING, turning, grinding, welding, shaping
Mounting	Usually lighter; may be moved/clamped by hand	Clamped to the machine table

Design principles: 3-2-1 (six-point) LOCATING to constrain the part fully; foolproofing (cannot be loaded wrong); quick clamping; clearance for chips.

Gauges (metrology)

- Gauges CHECK whether a dimension is within tolerance — they do not necessarily measure the exact value.
- **LIMIT (GO / NO-GO) gauges:** the GO end must enter/pass and the NO-GO end must NOT — parts between the limits are accepted.
- **TAYLOR'S PRINCIPLE:** the GO gauge checks the maximum-metal condition on ALL dimensions together; the NO-GO checks the minimum-metal condition one dimension at a time.

Gauge	Checks
PLUG	Holes (internal)
SNAP & RING	Shafts (external)
FEELER	Clearances — e.g., valve tappet (Unit 4, Topic 2a)
RADIUS and screw-PITCH (form)	Profiles
SLIP	Precision length standards
INDICATING (dial indicator/comparator)	Deviation readings

Gauges are made of hardened, wear-resistant steel and carry their own tolerance (gauge-maker's tolerance).

Practice MCQs

Q63. A JIG differs from a fixture in that it also:

(a) Holds the tool only
(b) Guides the cutting tool (via bushes)
(c) Measures parts
(d) Welds

Q64. A FIXTURE is typically used for:

(a) Drilling with guide bushes
(b) Milling, turning, welding (holds work, no tool guidance)
(c) Threading
(d) Marking

Q65. Jigs are most associated with which operation?

(a) Milling
(b) Turning
(c) Drilling/reaming/tapping
(d) Grinding

Q66. The location principle used to constrain a workpiece fully is:

(a) 3-2-1 (six-point)
(b) 1-1-1
(c) 5-5-5
(d) 2-2-2

Q67. A GO / NO-GO gauge is a type of:

(a) Indicating gauge
(b) Limit gauge
(c) Slip gauge
(d) Radius gauge

Q68. A plug gauge is used to check:

(a) Shaft diameters
(b) Hole (internal) diameters
(c) Thread pitch
(d) Radius

Q69. Snap and ring gauges check:

(a) Holes
(b) External (shaft) dimensions
(c) Angles
(d) Flatness

Q70. By Taylor's principle of gauging, the GO gauge checks:

(a) Minimum-metal, one dimension
(b) Maximum-metal condition on all dimensions together
(c) Only length
(d) Surface finish

Answers

Q63 — (b) Guides the cutting tool (via bushes) — *A JIG guides the tool (drill bushes).*
Q64 — (b) Milling, turning, welding (holds work, no tool guidance) — *A FIXTURE only holds (milling/turning).*
Q65 — (c) Drilling/reaming/tapping — *Jigs: drilling, reaming, tapping.*

Q66 — (a) 3-2-1 (six-point) — *3-2-1 six-point location.*

Q67 — (b) Limit gauge — *GO/NO-GO = limit gauge.*

Q68 — (b) Hole (internal) diameters — *Plug gauge — holes.*

Q69 — (b) External (shaft) dimensions — *Snap/ring gauges — shafts.*

Q70 — (b) Maximum-metal condition on all dimensions together — *Taylor's principle — GO checks max-metal on all dimensions; NO-GO checks min-metal one dimension at a time.*

Sources: Hajra Choudhury — Elements of Workshop Technology (Vols I & II); Kalpakjian — Manufacturing Engineering & Technology; standard production-engineering texts; NPTEL manufacturing lectures.
